Adaptive Feature Separation Network for Remote Sensing Object Detection

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***Abstract*— With the development of remote sensing technology, remote sensing object detection has been widely applied in various fields, but it still faces some thorny challenges, such as the following: 1) the complexity of object scale changes in remote sensing images makes it difficult to improve the performance of small object detection and 2) remote sensing images have complex backgrounds and densely arranged small and weak objects, which pose a serious problem of feature interference. To alleviate these challenges, we propose an end-to-end adaptive feature separation network called AFSNet, which includes a scale-aware module (SAM) and a class-aware module (CAM). The SAM mainly enables feature maps of different resolutions to detect objects of different scales. Shallow feature maps mainly suppress the features of large objects they contain to focus on small object detection, while deep feature maps increase the detailed features of large objects they contain to focus on large object detection. The CAM is mainly used to distinguish the features in the feature map by category, separating the features of different categories into different channels, thus mitigating the problem of inter class feature interference, and blocking background interference. The effectiveness of this article has been proven on the NWPU VHR-10, IPIU-M, DIOR, and DOTA2.0 datasets. It can be widely applied in civilian, military, and other fields. Through experimental verification, our AFSNet achieved 97.70% mAP on the NWPU VHR-10 dataset, 78.9% mAP on the DIOR dataset, and 58.22% mAP on the DOTA2.0 dataset. Our code is available at: https://github.com/Xidian-AIGroup190726/AFSNet.**

***Index Terms*— Deep learning, feature influence, multiscale, object detection, remote sensing images, small and weak objects.**

I. Introduction

OBJECT detection aims to detect the object of interest in a given image, not only recognizing the category of the object, but also accurately locating its position, which is a challenging task. With the increase in computing resources and the continuous improvement in deep neural network models [1], [2], the existing object detectors have achieved significant performance improvements, displaying quite satisfactory results, especially in natural images. However, these methods cannot achieve satisfactory performance on remote sensing images due to the special imaging principle of remote sensing images [3], [4], which is why remote sensing images are usually taken from satellites with a wider field of view. We further propose the analysis of remote sensing images as the motivation for our article, as shown in Fig. 1.

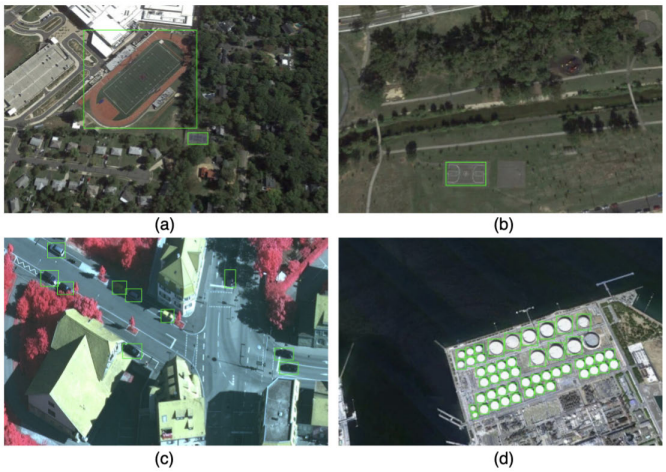


Fig. 1. Characteristics of remote sensing images. (a) Complex scale changes. (b) and (c) Interference between object features and background features. (d) Feature interference between dense objects.

1) *Complex Scale Variation:* In the same remote sensing scene, there is a significant difference in object scale, as shown in Fig. 1(a).
2) *Background Complexity:* Remote sensing images usually cover a vast geographical area, with complex scenes and backgrounds that are often similar to the foreground, which can easily interfere with the foreground, as shown in Fig. 1(b).
3) *Large Number of Small Objects:* Due to high-altitude photography, the pixel occupancy rate of most objects is low, and some objects only occupy a few dozen or even a few pixels, as shown in Fig. 1(c).
4) *Dense Object Arrangement:* In aerial images, objects are usually densely arranged, resulting in fuzzy intraclass boundaries and interference with interclass features, as shown in Fig. 1(d).

All of these have brought difficulties to the recognition and localization of objects in remote sensing images. Analyzing these characteristics can be summarized as two major challenges in remote sensing object detection: 1) the scale changes of objects in remote sensing images are complex, and

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it is difficult to balance objects of different scales, especially small objects and 2) remote sensing images have complex backgrounds and small objects are often densely arranged, leading to serious feature interference problems.

For Challenge 1, which is the complexity of scale changes in remote sensing images, researchers have proposed many multi-scale object detection works for remote sensing images. mSODANet [5] introduces a hierarchical expansion network on top of the multiscale features extracted from the backbone network, effectively learning contextual information of various objects in multiscale and multifield of view. mSODANet also utilizes a bidirectional feature aggregation module to process the multiscale contextual information extracted from the hierarchical expansion network multiple times, establishing effective multiscale contextual information in the network and further enhancing object detection capabilities. GLSANet [6] proposes a new global-to-local scale-aware detection network, which mines and enhances advanced semantic information in deep feature maps by designing a global semantic interaction module to reduce background interference on foreground objects. In addition, GLSANet adopts an optimized FPN to integrate enhanced semantic information into multiscale feature maps, achieving multiscale object detection. Finally, GLSANet also highlights the features of small objects by introducing local attention pyramids, and suppresses background and noise without additional complex filtering methods.

However, most methods first obtain higher level features through different methods, and then enrich the backbone features through feature aggregation, without considering that the features of large objects have higher responses. These high responses weaken the only feature representation of small objects and hinder their detection.

For Challenge 2, which is difficult to detect weak small objects in complex remote sensing images due to feature interference, many researchers have proposed the idea of denoising images and feature maps. Cho et al. [7] have constructed a deep denoising autoencoder model to remove adversarial disturbances and restore clean images. Tian et al. [8] utilize a sparse module, feature enhancement module (FEM), attention module, and reconstruction module to extract salient features that integrate global and local features, thereby removing noise hidden in complex backgrounds and reconstructing noise-free images. Xie et al. [9] add denoising blocks in the middle layer of convolutional networks, and use adversarial training to jointly train denoising blocks with each layer of the network end-to-end, enabling the network to eliminate feature map noise that partially depends on data. However, few jobs are specifically designed for object detection tasks.

To address the aforementioned challenges, we propose the adaptive feature separation network (AFSNet), with specific contributions as follows.

1) For the complex scale changes in remote sensing images, we did not introduce additional high-level features for feature fusion and enhancement. Instead, we proposed a feature highlighting and enhancement module (SAM) for scale perception based on the essence of different resolution feature maps. Due to the small receptive field of shallow feature maps, most of the extracted features are detailed features, mainly used for detecting small objects, while deep feature maps have a large receptive field and extract more semantic features, mainly used for detecting large objects. Therefore, this module activates objects of different scales on different predicted feature maps of the network, highlighting small object areas on low-level feature maps and enhancing detailed features of large objects on deep feature maps. Different resolution feature maps each play their own role, thus avoiding the mutual influence of objects of different scales on the same resolution feature map and resulting in low detection performance.
2) For the problem of weak objects being difficult to detect due to feature interference, we did not focus on important object features through attention or extract and enhance semantic information through some additional methods. We fully tap into the potential of the information contained in the feature map itself, focusing on utilizing the inherent feature information of the feature map, and propose a class-aware module (CAM). By separating all features in the predicted feature map according to category in the channel dimension, and separating features of different categories into different channels, we obtain a feature map with clear boundaries, reducing feature interference between densely arranged objects. At the same time, we also separate the background features into corresponding channels, effectively reducing the interference of complex background features on weak object features.

II. Related Work

First, we discuss some current multiscale object detection methods and also present some representative work on feature denoising.

A. Multiscale Object Detection

The objects in remote sensing images may have different scales and sizes, as they can be located in different geographical locations and captured at different heights. By using multiscale detection, the model can adapt to objects of different sizes, improve its robustness to objects, and ensure that the model can effectively detect objects of different sizes.

Multiscale object detection has always been a research hotspot for researchers, aiming to improve the detection performance of objects at different scales. At the beginning, researchers introduced the concept of image pyramid [10], [11], which allows the model to perceive object information more comprehensively and improve the detection accuracy of small objects by inputting images of different resolutions. Subsequently, in response to the limitations of traditional single-scale feature extraction methods [12], researchers began to focus on designing complex multiscale feature extraction networks [13], [14], [15], which can effectively extract multilevel feature representations from input images, provide richer information for detectors, and enhance the model's detection ability for various objects. In recent years, researchers have attempted to introduce attention mechanisms into object

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detection networks [16], [17], [18], which adaptively focus on important information at different scales, further optimizing the model's perception of multiscale objects and improving detection performance.

Although these algorithms have improved overall detection performance through different methods, most detectors have overlooked a problem: the rich features of shallow large objects will guide the network to pay more attention to them, and the training network will gradually ignore the already scarce small object features, resulting in poor small object detection performance.

In this work, we propose a feature highlighting and enhancement module for scale perception, aiming to highlight the regional features of small objects that should be detected in the current layer for low-level high-resolution feature maps, suppress the features of medium and large objects that do not correspond to the corresponding scale in the current layer, and enhance the detailed features of large objects that should be detected in the current layer for deep low-resolution feature maps, in order to detect large objects. In this way, different layers only have a higher response to the object area of the corresponding scale that should be detected, thus achieving more accurate object detection.

B. Feature Denoising

In the field of object detection, the research on feature denoising aims to improve the model's effective perception of the object and reduce interference from irrelevant features in the feature map. Early work mainly focused on introducing feature selection and attention mechanisms [19], so that the model can selectively focus on regions related to the object, reduce the influence of irrelevant information, and improve the local perceptual features of the object. Later, GANs were also used for feature denoising [20]. By training a generator network, the model can learn to remove noise from feature maps, improve the quality of features, and facilitate accurate object detection. At present, more and more researchers are also aware of the impact of noise on object domain detection performance in transfer learning and unsupervised adaptive work. RFA-Net [21] proposes the SFR module to reconstruct low-level features, and uses low-rank matrix reconstruction to denoise redundant features, thereby eliminating the interference of background on features. Biswas and Tešić [22] propose a novel pseudo-labeling technique that relies on N-shot learning to remove unwanted background noise from object proposals, significantly improving object domain detection performance.

However, most of the above methods rely on external methods such as attention mechanisms or GANs to remove background noise in feature maps. They overlook the potential inherent in the feature map itself. In this work, we propose a category-aware feature separation module that decouples feature maps based on categories. Different categories of features and background features are decoupled into different channels, blocking background interference while effectively reducing interclass interference caused by close arrangement of objects in different categories.

C. Small Object Detection

Small object detection has always been a challenging issue in the field of object detection. Due to the small object itself having fewer pixels, after multiple downsampling, the small object features on the predicted feature map become more scarce, resulting in lower small object detection performance. Therefore, in recent years, many researchers have been committed to improving the performance of small object detection.

Lee et al. [23] bridged the connection between the primary and secondary feature pyramid networks by introducing attention modules and sigmoid functions, and further refined the RFP output by using sigmoid functions to filter out background noise, effectively amplifying the spatial detail features of small objects and improving their detection accuracy. SOLO-Net [24] proposed a sparse outlook attention module and a sparse outlook path aggregation network as the fundamental component of the network, effectively improving the performance of the path aggregation network and enhancing the model's ability to focus on small objects. Second, SOLO-Net proposed a sigmoid intersection over union loss function specifically designed for small objects, which effectively improves the detection performance of small objects. VHNet [25] establishes a vertical and horizontal cross-information flow within the neural network, which can fuse low-level and high-level features as well as features of different sizes, thereby improving the accuracy of small object detection. RCFNet [26] proposes a new small object detection method based on radar vision fusion. For the extracted features of both radar and camera modalities, attention is first used to align the image semantic information with radar features. Then, cross transformer is used to deeply fuse the two modalities in a global manner, fully utilizing the data information of radar and camera, effectively improving the performance of small object detection on water surfaces. Gao et al. [27] proposed a feature super-resolution fusion framework based on cross-scale distillation and designed a subpixel super-resolution feature pyramid network (SSRFPN), which uses subpixel super-resolution (SSR) modules instead of bilinear interpolation to enhance feature expression ability. In addition, SSRFPN proposes a cross-scale distillation mechanism to guide the SSR module to learn the features of small object regions more accurately, effectively improving the detection performance of small objects.

The above methods mainly focus on the regional features of small objects through attention mechanisms, or assist detection by adding contextual information of small objects through feature fusion. However, this article found that in remote sensing images, small objects are often easily overwhelmed by the high response of surrounding large objects or influenced by complex backgrounds. Therefore, starting from eliminating the interference of small object features, this article effectively highlights the features of small objects by suppressing the high response features of large objects and blocking complex background features, thereby improving the detection performance of small objects.

III. Method

Due to the special imaging mode, remote sensing images present characteristics such as excessive object scale

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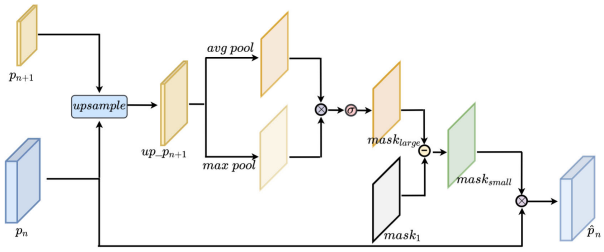


Fig. 2. Structural flowchart of the FHM module. First, the rough distribution $\text{mask}_{\text{large}}$ of the large object is calculated using the deep feature map p_{n+1}, and the $\text{mask}_{\text{large}}$ is inverted to obtain the distribution $\text{mask}_{\text{small}}$ of the small object. By applying it to the low-level feature map p_n, the characteristics of the small object in the low-level feature map are highlighted, and the characteristics of the large object are suppressed.

differences and complex terrain. In order to solve the problem of multiscale detection caused by excessive object scale differences in remote sensing image object detection and the problem that objects of each category easily interfere with each other in complex scenes, we have designed a novel object detection model based on adaptive feature separation, which can achieve good accuracy in remote sensing images containing multiscale objects and complex terrain.

A. Scale-Aware Feature Highlighting and Enhancement Module

Due to the presence of objects of different scales in a remote sensing image, with the superposition of training times, the network will tend to focus more on the features of large objects with larger contributions, and the regions with significant small objects will be gradually weakened or even ignored. Therefore, it is difficult to simultaneously detect objects of different scales in feature maps of the same scale. Thus, this article proposes a scale-aware feature highlighting and enhancement module. This module can be divided into two submodules: feature highlighting module (FHM) and FEM.

1) Feature Highlighting Module: This module is mainly used to suppress the influence of high response of large objects on small objects in low-level feature maps. The module structure is shown in Fig. 2.

Assuming that p_n is the input low-level feature map and p_{n+1} is the input high-level feature map, we first use the high-resolution feature map $\text{up_}p_{n+1}$ of p_{n+1} after nearest-neighbor upsampling to obtain the rough location of the large object

$$\text{mask}_{\text{large}} = \sigma(\mathbf{A}(\text{up_}p_{n+1}) \times \mathbf{M}(\text{up_}p_{n+1})) \tag{1}$$

where $\mathbf{A}(\cdot)$ and $\mathbf{M}(\cdot)$ denote the average pooling and max pooling, respectively. $\sigma(\cdot)$ denotes the sigmoid function. Because the high-level feature map loses a lot of spatial features of small objects after a large number of convolutional pooling operations and retains more features of large objects, the sigmoid function can be used to activate the high response of large objects and obtain the response region of large objects.

Then using $\text{mask}_{\text{large}}$ we can calculate the rough position of the small objects

$$\text{mask}_{\text{small}} = \text{mask}_1 - \text{mask}_{\text{large}} \tag{2}$$

where mask_1 represents the feature map with the same resolution size as $\text{mask}_{\text{large}}$ and the value is all 1. Due to the fact that the value of $\text{mask}_{\text{large}}$ after the sigmoid function falls within the range of (0, 1), and the position response of the large object is high, the value approaches 1. Therefore, we can use mask_1 to subtract $\text{mask}_{\text{large}}$ to obtain its complementary feature map, which can obtain a rough response area of the small object.

Therefore, the low-level feature maps that are suppressed with large object features can be obtained by the following calculation:

$$\begin{aligned}\hat{p}_n &= p_n - p_n^{\text{large}} \\ &= p_n - \text{mask}_{\text{large}} \times p_n \\ &= p_n \times \left(1 - \text{mask}_{\text{large}}\right) \\ &= p_n \times \text{mask}_{\text{small}}\end{aligned} \tag{3}$$

where p_n^{large} denotes the features of the large object area that needs to be suppressed in p_n obtained through p_{n+1}.

So far, we have calculated the low-level feature map $\hat{p}_n$ after feature highlighting, and ideally, $\hat{p}_n$ should suppress the high response of the large object and highlight only the regional features of the small object, and the detection results should be better. However, experiments have revealed that the detection accuracy as well as recall of such an approach has been reduced rather than improved. We suspect that this is caused by the ineffective propagation of coarse location information of large objects from high-level to low-level layers. Therefore, we visualized the output results $\hat{p}_n$ of FHM, as shown in Fig. 3(b), and found that although the features of large objects were suppressed in the low-level feature map, some useful small object features are also eliminated. This is because we did not consider the semantic differences between feature maps of different resolutions when we utilized them, and the rough nearest-neighbor upsampling led to the misalignment of the features between the different feature layers, which eliminated some useful features in the low-level feature maps p_n.

To alleviate this problem, we are inspired by the concept of optical flow, which is widely used in video processing tasks, and propose to learn the semantic flow between feature maps of different resolutions to guide deep feature maps for upsampling. Optical flow is used to represent the apparent motion patterns of objects, surfaces, and edges in a visual scene induced by relative motion, and it is very effective and flexible in aligning two neighboring video frame features in video processing tasks. The idea of optical flow motivated us to design a flow-based offset module, where we represent the relationship between two feature maps of different resolutions of the same image in terms of the "motion" of each pixel from one feature map to the other, and align two neighboring levels of feature maps by predicting the flow field inside the network. We define such a flow field as a semantic flow, which is generated between different levels of the feature pyramid. In this way, once the precise semantic flow is obtained, the network is able to propagate the semantic features with minimal information loss, thereby better transmitting the response area of large objects to low-level layers. Its specific structure is shown in Fig. 4.

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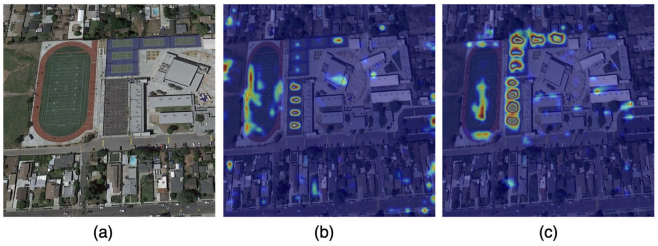


Fig. 3. Visualization of SAM processed feature maps without offset and with offset upsampling. (a) Input image. (b) Features of some small objects are suppressed due to feature misalignment. (c) Large object features in the shallow feature map are suppressed, while the small object features are well highlighted.

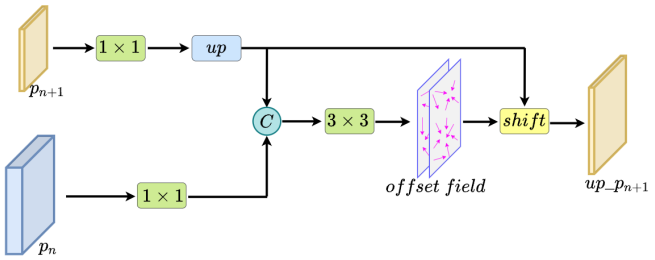


Fig. 4. Upsampling flowchart with offset. Guide upsampling of deep feature maps p_{n+1} by learning the semantic flow offset field between feature maps of different resolutions.

The calculation process can be expressed by the following equation:

$$\text{offset field} = \mathbf{Conv}_3(\mathbf{cat}(\mathbf{u_n}(\mathbf{Conv}_1(p_{n+1})), \mathbf{Conv}_1(p_n))) \quad (4)$$

$$\text{up_}p_{n+1} = \mathbf{shift}(p_{n+1},\ \text{offset field}) \quad (5)$$

where offset field is the upsampling offset field learned from feature maps of different resolution sizes, used to mitigate feature misalignment between different feature layers. $\mathbf{cat}(\cdot)$ represents the concat operation of the channel dimension, and $\mathbf{u_n}(\cdot)$ is the well-known nearest-neighbor upsampling. $\mathbf{shift}(\cdot)$ is a shift transformation operation that uses offset field to calculate the sampled feature map on a low-resolution feature map. The specific operation is shown in Fig. 5. When calculating the feature values at p_1 in the upsampling feature map up_p_{n+1}, in order to obtain more accurate sampling points, we first find the offset off_{p_0} of p_0 in the original feature map p_{n+1} at the corresponding position of offset field, apply it to point p_0, and then offset to obtain p_2. Therefore, the feature value at p_2 in p_{n+1} is the feature value at p_1 in up_p_{n+1}.

It is important to note that semantic flow is clearly different from optical flow because semantic flow takes input feature maps at different levels and evaluates the differences between them to find a suitable flow field that will dynamically indicate how to effectively align the two feature maps. Therefore, by clearly and dynamically establishing the positional correspondence between feature maps, we have effectively solved the problem of their actual misalignment during the interaction process.

Therefore, we replace the upsample operation in Fig. 2 with this module and visualize the final output of the low-level feature map, as shown in Fig. 3, which can be seen that it only eliminates the features of the large object, thus highlighting the features of the small object very well.

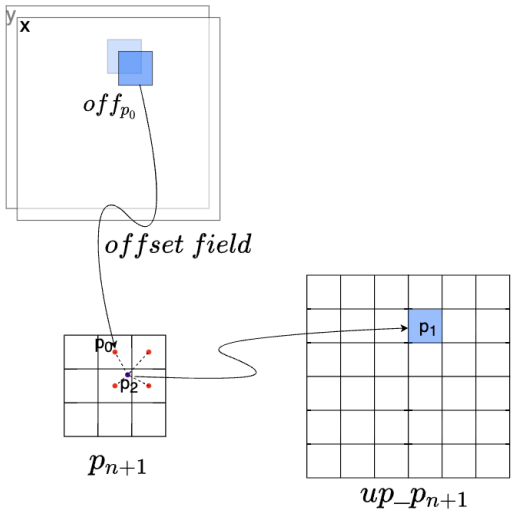


Fig. 5. Upsampling with offset: the corresponding point after the upsampling of p_0 is p_1. p_0 first finds the offset at p_1 in the offset field, then uses the offset to calculate the position of the real sampling point p_2, and finally uses the values of the four points around p_2 to perform bilinear interpolation to obtain the value at p_1.

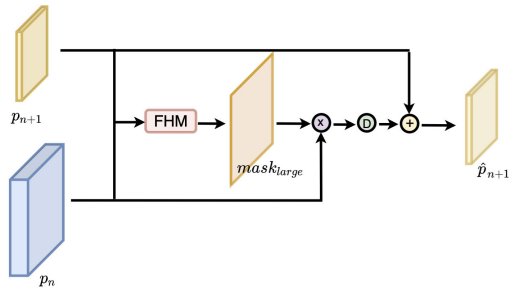


Fig. 6. FHM can obtain the $\text{mask}_{\text{large}}$ of the large object, which is multiplied by the shallow feature map to obtain the detailed features of the large object. Then, it is downsampled and added to the original deep feature map p_{n+1} to obtain the enhanced deep feature map $\hat{p}_{n+1}$.

2) Feature Enhancement Module: This module is mainly used to enhance the detailed features of large objects in high-level feature maps, and its module structure is shown in Fig. 6. The output feature map $\hat{p}_{n+1}$ can be obtained as follows:

$$\begin{aligned}\hat{p}_{n+1} &= p_{n+1} + p_n^{\text{large}} \\ &= p_{n+1} + \mathbf{D}\left(\text{mask}_{\text{large}} \times p_n\right)\end{aligned} \quad (6)$$

where p_n^{large} represents the detail features of the large objects in the feature map p_n. $\mathbf{D}(\cdot)$ represents the downsampling operation, and $\text{mask}_{\text{large}}$ is the rough location of the large objects obtained in the FHM module. Because the high-level feature map with large convolution and pooling layers has a large receptive field, more semantic information about the large objects is extracted, while detailed information is scarce. Therefore, using $\text{mask}_{\text{large}}$ to capture the detailed features of the large objects in the low-level feature map, fully integrating the semantic and detailed features of large objects, and effectively enriching the feature information of large objects, which is also more conducive to the detection of large objects.

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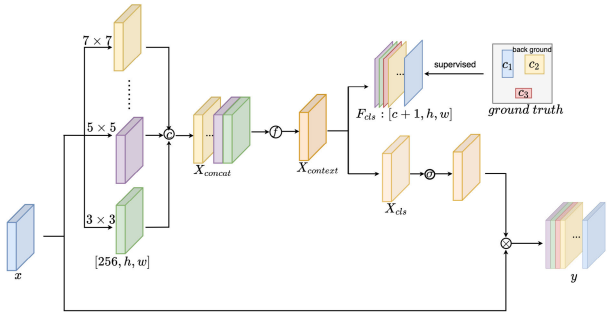


Fig. 7. Input feature map x first obtains more contextual information through multiple convolutional layers with different kernel sizes, and then uses concat and another convolutional layer $f(\cdot)$ to fuse the captured contextual features. After that, X_{context} is fed into two parallel branches, one of which generates a feature map F_{cls} with a channel number of $c+1$ under the supervision of ground truth, c is the number of class, and the other branch generates X_{cls} under the indirect influence of F_{cls} to calculate category separation weights, then multiply with the original feature map x to obtain a feature map y with category separation.

In summary, our SAM detects objects of different scales in different layers by establishing relationships between feature maps at different scales, increasing the network's ability to perceive scales and enabling efficient multiscale detection.

B. Class-Aware Feature Separation Module

Due to the frequent small and dense arrangement of objects in remote sensing images, the boundaries of objects in the feature map are unclear, resulting in interference between objects. Therefore, we propose a feature separation module for category perception, which is mainly aimed at reducing mutual interference between different category features and the impact of complex backgrounds on objects. The specific structure is shown in Fig. 7.

Suppose x is the input feature map, then

$$X_{\text{context}} = \mathbf{f}(X_{\text{concat}}) = \mathbf{f}\left(\bigcup_{i=1}^{n} \mathbf{Conv}_{k_i}(x)\right) \tag{7}$$

$$F_{\text{cls}} = \mathbf{conv}_1(X_{\text{context}}) \tag{8}$$

$$\text{score}_{\text{sep}} = \mathbf{conv}_1(X_{\text{context}}) \tag{9}$$

where X_{context} is a feature map with sufficient contextual information obtained by convolutional layers with different convolutional kernel sizes, which is shared by F_{cls} and $\text{score}_{\text{sep}}$ branches, and $\mathbf{f}(\cdot)$ is the 1×1 convolutional layer used to further fuse the acquired contextual features. F_{cls} is a category response map predicted under the supervision of category ground-truth map, with a total number of categories $\text{num}_{\text{cls}}+1$ channels, whose purpose is to guide X_{context} to gradually become sensitive to categories during the training process, thereby generating a separation score $\text{score}_{\text{sep}}$ for separating feature maps X_{context}. $\text{score}_{\text{sep}}$ is obtained by the convolution of X_{context} under the indirect influence of F_{cls}.

Therefore, our output feature map Y can be obtained as follows:

$$Y = \sigma\left(\text{score}_{\text{sep}}\right) \odot X = W_c \odot X = \bigcup_{i=0}^{\text{cls}} W_{c_i} \odot X^i \tag{10}$$

where $\sigma(\cdot)$ denotes the sigmoid function, which scales the separation score to between 0 and 1. The obtained W_c is used as the 3-D weight of the input feature map based on category, space, and channel, which separates the input feature map based on category, so it can be further split into W_{c_i}. $\bigcup_{i=0}^{\text{cls}}$ represents the concatenation of categories and background features in the channel dimension, while $i = 0$ represents the background. W_{c_i} represents the corresponding weight of category i at each position, X^i represents the features of category i. This is more like adding a category attention on the basis of spatial attention and channel attention, and finally, this weight is multiplied element-by-element with the input feature map to obtain the output feature map with clear category boundaries by weighting the features of different categories.

However, for a feature map, its content is complex, so we have divided it into more detailed categories based on feature attributes. Therefore, the above equation can be further expressed as follows:

$$\begin{aligned} Y &= \bigcup_{i=0}^{\text{cls}} W_{c_i} \odot X^i \\ &= \underbrace{W^{\text{bg}} \odot X^{\text{bg}}}_{\text{background}} \cup \underbrace{\bigcup_{j=1}^{\text{cls}_{\text{in}}} W^j \odot X^j}_{\text{categories in image}} \cup \underbrace{\bigcup_{k=1}^{\text{cls}_{\text{not}}} W^k \odot X^k}_{\text{categories not in image}} \\ &= \underbrace{\bigcup_{m=1}^{C_{bg}} w_m^{bg} \odot x_m^{bg}}_{\text{background}} \cup \underbrace{\bigcup_{j=1}^{\text{cls}_{\text{in}}} \bigcup_{n=1}^{C_j} w_n^j \odot x_n^j}_{\text{categories in image}} \cup \underbrace{\bigcup_{k=1}^{\text{cls}_{\text{not}}} \bigcup_{l=1}^{C_k} w_l^k \odot x_l^k}_{\text{categories not in image}} \end{aligned} \tag{11}$$

where X^{bg} represents the background features in the image, X^j represents various types of features that appear in the image, X^k represents a small number of features introduced in the training and learning process that do not appear in the image, W^{bg}, W^j, and W^k correspond to the weights of each part respectively, and cls_{in} and cls_{not} represent the number of categories that appear in the image and the number of categories that do not appear in the image, respectively. Due to the complexity of background features and the introduction of additional category features, it is inevitable that they will bring some interference to our object features. Therefore, we can effectively reduce interclass and background interference by using weights to weigh different parts of the features, increasing the weight of category features that appear in the image, and reducing the weight of background and nonappearing categories.

Because different categories contain features with different attributes and occupy different numbers of channels, we can further weight the features based on channels. C_{bg}, C_j, and C_k represent the number of channels contained in each category's features, w_n^j represents the weight of category j at various positions in the nth dimensional channel, and x_n^j represents the corresponding features of category j's nth dimensional channel, so even if it is an object of the same category, after more detailed weighting of channel dimensions, the object

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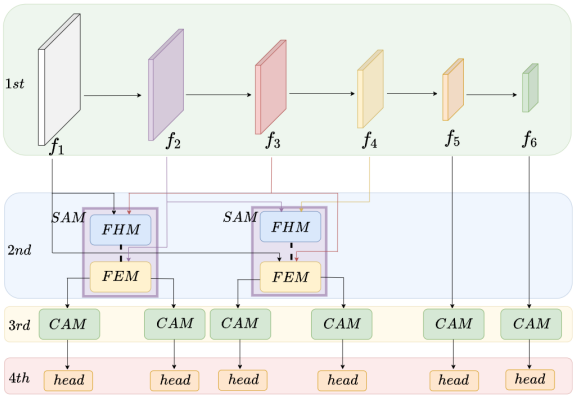


Fig. 8. Overall network of the AFSNet. f_1–f_6 represent six different resolution feature maps extracted by SSD. The two discontinuous feature maps f_1 and f_3, f_2 and f_4 are respectively fed into two SAMs, FHM is used to highlight the features of low-level small objects, and FEM is used to enhance the detailed features of deep feature maps of large objects. The four feature maps output by SAM and f_5 and f_6 are then subjected to a CAM to obtain the final six predicted feature maps for object detection.

features will gradually be separated into different channels, making the object boundaries clearer.

In summary, our CAM separates the feature maps extracted from the network into 3 dimensions: category, space, and channel. Features of different categories are separated into their respective channels, and each channel displays a feature of a category, making the features clearer and clearer, effectively reducing the impact of mutual interference between objects on detection.

C. Overall Network

Our AFSNet network is shown in Fig. 8, which mainly consists of four stages: feature extraction backbone, scale-aware feature highlighting and enhancement, class-aware feature separation, and detection head.

1) Feature Extraction Backbone: In the first stage, the input image is processed through SSD network to extract different scales of feature maps [f_1, f_2, f_3, f_4, f_5, f_6]. The low-level feature maps have high resolution, and the extracted features are more detailed features of the object, while the high-level feature maps have low resolution, and the extracted features are more semantic features of the object.

2) Scale-Aware Feature Highlighting and Enhancement Module: The input of this stage is feature maps extracted in the first stage, and a low-level feature map and a high-level feature map are selected and fed to this stage, respectively. The low-level feature map uses the high-level feature map to suppress the strong responses of medium and large objects in the low-level layer, and the high-level feature map uses the low-level feature map to obtain the detailed features of medium and large objects. The output of this stage is a low-level feature map with the small objects highlighted and a high-level feature map with the detailed features enhanced, and the feature map size is kept consistent with the input.

3) Class-Aware Feature Separation Module: This module mainly separates the complex feature maps into different channels according to the categories to avoid the features between complex objects interfering with each other. The feature maps processed in the second stage are fed into this module separately, and the output feature maps with exactly the same input shape are obtained for the last stage of detection.

4) Detection Head: The detection head receives the predicted feature map output from the previous stage for classification and regression and outputs the predicted category score and the corresponding offset for each default box. Based on the predicted category score and the offset of the default box, we can calculate the coordinates and categories of the final predicted box.

5) Loss Function: Our loss function is a multitask loss function. The total loss contains three components: classification loss, regression loss, and category feature separation loss, and is formulated as follows:

$$\begin{aligned}\mathbf{L} &= \alpha\mathbf{L}_{\mathbf{reg}} + \beta\mathbf{L}_{\mathbf{cls}} + \gamma\mathbf{L}_{\mathbf{cfs}} \\ &= \frac{\alpha}{N}\sum_{n=1}^{N} t'_n \cdot \sum_{j\in\{x,y,w,h\}} l_{\text{reg}}\left(p_{nj}, t_{nj}\right) \\ &\quad + \frac{\beta}{N}\sum_{n=1}^{N} l_{cls}(p_n, t_n) \\ &\quad + \gamma\sum_{f=1}^{m}\frac{1}{h_f \times w_f}\sum_{i}^{h_f}\sum_{j}^{w_f} l_{\text{cfs}}\left(p_{ij}, t_{ij}\right) \end{aligned} \tag{12}$$

where α, β, and γ denote the hyperparameters for balancing the three loss components, respectively, and are set to 1 by default at initialization.

a) $\mathbf{L}_{\mathbf{reg}}$*:* We use the classical smooth $L1$ loss to calculate the regression loss of the predicted boxes, where N denotes the number of default boxes, t'_n is a binary variable with value 0 or 1, p_{nj} denotes the offset of the predicted default boxes, and t_{nj} denotes the offset between the ground-truth boxes and the default boxes. t'_n is mainly used to control the regression loss of positive and negative samples. If the current default boxes match the real object, it is a positive sample, $t'_n = 1$, if the current default boxes do not match the real object, it is a negative sample, $t'_n = 0$, that is, we only calculate the regression loss of positive samples, the regression loss of negative samples is 0.

b) $\mathbf{L}_{\mathbf{cls}}$*:* We use focal loss to calculate the classification loss of each prediction box. p_n is the probability that the current default boxes are predicted to be of each category, and t_n is the true category to which the current default boxes belong.

c) $\mathbf{L}_{\mathbf{cfs}}$*:* According to the idea of separating complex feature maps by category, we need to predict the category response map F_{cls} so that the feature map has a certain perceptual ability to the category, and then generates the category separation weights by the feature map, from which we can see that the more accurate the F_{cls}, the stronger the perception of the feature map to the category, the clearer the new feature map obtained after separation, and the less interference between categories, so we propose the loss for supervised generation of F_{cls}, which is equivalent to a pixel-level classification loss, the specific loss function used is softmax cross-entropy, p_{ij}

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denotes the predicted probability distribution of the category to which the pixel point (x, y) belongs, and t_{ij} denotes the true category to which the object at the pixel point (x, y) belongs. This design can make the category response graph F_{cls} more approximate to the true category distribution, and the F_{cls} is more sensitive to categories, thus possessing more accurate category response-ability.

IV. Experiments

In this section, we first briefly introduce the dataset used in our experiments. Then, specific experimental parameter configurations are provided. Finally, we demonstrate the effectiveness of each of our proposed modules through a series of ablation experiments and compare them with the existing object detection algorithms on our datasets.

A. Dataset Description

In this article, we use three datasets to validate our method. They are NWPU [28], IPIU-M, and DOTA2.0, respectively. In addition, we conducted comparative experiments on the DIOR dataset.

NWPU is a challenging ten-class geospatial object detection dataset released by NPU in 2014. The ten categories of objects are airplane, ship, storage tank, baseball diamond, tennis court, basketball court, ground track field, harbor, bridge, and vehicle. The dataset contains a total of 800 VHR optical remote sensing images, including 715 color images from Google Earth with a spatial resolution of 0.5–2 m, and 85 pan-sharpened color infrared images from Vaihingen data with a spatial resolution of 0.08 m. The dataset contains two image sets, a positive image set containing 650 images, each containing at least one object, and the number of objects labeled in this image set is shown in Table I; the other is a negative image set with 150 images that do not contain any objects, which is mainly used for object detection based on semisupervised learning and object detection based on weakly supervised learning. In this dataset, categories such as vehicles, and basketball courts are usually included in complex background environments, and storage tanks are usually arranged densely, which are just good features to verify the effectiveness of our module.

Another dataset is IPIU-M, which has a spatial resolution from 0.15 to 1.2 m and contains 1739 images acquired from Google Earth and Baidu Maps. The number of annotated objects of each category in this dataset is shown in Table II, with a total of 24 513 instances annotated on four categories. In addition, the large proportion of small and dense objects in the IPIU-M dataset makes the detection task more challenging, because the detection of small and dense objects has been a major challenge in the field of detection, so the IPIU-M dataset is also just right to verify the robustness of our proposed model to small objects.

DIOR is a large-scale benchmark dataset used for object detection in optical remote sensing images, consisting of 23 463 images and 190 288 object instances, with a spatial resolution of 0.5–30 m. The image size in the dataset is 800 × 800 pixels, covering 20 object classes. These 20 object classes include airplane (AL), airport (AT), baseball field (BF), basketball court (BC), bridge (BG), chimney (CM), dam (DM), expressway-service-area (EA), expressway-toll-station (ES), vehicle (VH), harbor (HB), overpass (OP), golf course (GC), ground track field (GF), ship (SP), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), and windmill (WM).

DOTA2.0 dataset comprises 2430 overhead images sourced from Google Earth, Gaofen-2 (GF-2), and Jilin-1 (JL-1) satellites. The GSD in the dataset ranges from 0.1 to 0.87 m. DOTA v2.0 has 18 categories including plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), helicopter (HC), container crane (CC), airport (AP), and helipad (HL).

B. Experimental Setup

Our model baseline adopts an SSD detection algorithm, and we use VGG16 [29] to extract features from the image. However, we did not use all layers of VGG16, only the first four blocks were used for feature extraction. Then, we added a series of additional layers after the output feature map of the last block, and finally selected six effective feature layers. Then, we added the module we proposed in this work to these six prediction feature maps for the final prediction.

The specific training hyperparameters are shown in Table III. In order to cover as many sizes of objects as possible, we define the size of default boxes on each predicted feature layer by minSize and maxSize as in SSD, and the aspect ratio of default boxes on the corresponding layer. In addition, to avoid the feature map being too small, we resize the input image to 400 pixels.

After superparametric experiments, we found that the scale-aware feature highlighting and enhancement modules work better on the first four feature maps only. We performed the class-aware feature separation module on all six predicted feature maps.

C. Comparison and Analysis of Hyperparameters

In this section, we provide a detailed comparison and analysis of the hyperparameters involved in the network. Hyperparameters in the model include the number **n** of scale-aware module (SAM) (scale-aware feature highlighting and enhancement modules), the weight α for categorization loss, β for regression loss, and γ for category separation loss in the total loss. We use average precision (AP) to evaluate the performance of object detection.

In our experiments, we use consistent parameters other than the hyperparameters that need to be determined for training and testing on each dataset. We used the fixed-variable method to determine the appropriate hyperparameters. We optimize the hyperparameters on the NWPU VHR-10 dataset, the IPIU-M dataset, the DIOR, and the DOTA2.0 dataset, respectively.

1) Hyperparameters Analysis for n: For the SSD object detection baseline network, there are six prediction feature layers, and we can choose any two of these six feature layers

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TABLE I
DIFFERENT CATEGORY OBJECT NUMBERS IN NWPU VHR-10

cls_name	airplane	baseball diamond	basketball court	bridge	ground track field
instance_number	757	390	159	124	163
cls_name	harbor	ship	storage tank	tennis court	vehicle
instance_number	224	302	655	524	477

TABLE II
DIFFERENT CATEGORY OBJECT NUMBERS IN IPIU-M

cls_name	helicopter	Boeing	dense ship	sparse ship
instance_number	727	672	18697	4417

TABLE III
TRAINING PARAMETERS

parameters	values
Input Image Size	400x400
anchor min_size	[25, 65, 116, 167, 218, 269]
anchor max_size	[65, 116, 167, 218, 269, 320]
aspect_ratios	[1, 2, 3]
Leaining Rate	0.0001
Momentum	0.9
Weight Decay	0.0005
Weight Initilization	Xaiver
Max Iteration	28500

TABLE IV
EXPERIMENTAL RESULTS OF HYPERPARAMETER n ON NWPU VHR-10 (%)

hyper-paramers		mAP
skip_sam = no	n=1	96.77
skip_sam = no	n=2	96.52
skip_sam = no	n=3	96.08
skip_sam = yes	n=1	96.99
skip_sam = yes	n=2	**97.70**
skip_sam = yes	n=3	96.73

TABLE V
EXPERIMENTAL RESULTS OF HYPERPARAMETER n ON IPIU-M (%)

hyper-paramers		mAP
skip_sam = no	n=1	76.70
skip_sam = no	n=2	76.34
skip_sam = no	n=3	75.98
skip_sam = yes	n=1	76.84
skip_sam = yes	n=2	**77.27**
skip_sam = yes	n=3	76.68

to perform a scale-aware feature highlighting and enhancement module; specifically, we experimented with adding scale-aware feature highlighting and enhancement modules for two consecutive prediction feature maps and for cross-layer prediction feature maps, as well as experimenting with the most appropriate number of such modules to be added to the overall network, as shown in Tables IV–VI.

2) Hyperparameters Analysis for α, β, and γ: We use different hyperparameters α, β, and γ to evaluate the effectiveness of the total loss function. As can be seen from Tables VII–X, our model improves over the baseline for different parameters, and the best performance is achieved when using the NWPU VHR-10 dataset when $\alpha = 0.5$, $\beta = 0.3$, and $\gamma = 0.3$. The best performance was achieved when the IPIU-M dataset was used with $\alpha = 0.5$, $\beta = 0.3$, and $\gamma = 0.5$. The best performance was achieved when the DIOR dataset was used with $\alpha = 0.5$, $\beta = 0.3$, and $\gamma = 0.2$. The best performance was achieved when the DOTA2.0 dataset was used with $\alpha = 0.6$, $\beta = 0.4$, and $\gamma = 0.3$. The reason for the different γ values in the different datasets is that different datasets have different numbers of categories. For example, the NWPU VHR-10 dataset has a total of ten categories, and in the class-aware feature separation module, the NWPU VHR-10 dataset needs to regress the 256-D feature map into $10 + 1$ channels for category perception, while IPIU-M only needs to regress into $4 + 1$ channels for category perception, and more categories lead to larger category perception loss values. This makes the model pay more attention to the category perception error during the learning process, which affects the performance of the model. Therefore, a smaller weight γ in the NWPU VHR-10 dataset allows the model to get a better balance between the loss of each module.

TABLE VI
EXPERIMENTAL RESULTS OF HYPERPARAMETER n ON DIOR (%)

hyper-paramers		mAP
skip_sam = no	n=1	78.2
skip_sam = no	n=2	78.1
skip_sam = no	n=3	77.9
skip_sam = yes	n=1	78.6
skip_sam = yes	n=2	**78.9**
skip_sam = yes	n=3	78.5

TABLE VII
EXPERIMENTAL RESULTS OF HYPERPARAMETERS α, β, AND γ ON NWPU VHR-10 (%)

hyper-paramers			mAP
$\alpha = 0.4$	$\beta = 0.3$	$\gamma = 0.3$	97.38
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.3$	**97.70**
$\alpha = 0.6$	$\beta = 0.3$	$\gamma = 0.3$	97.43
$\alpha = 0.5$	$\beta = 0.2$	$\gamma = 0.3$	97.23
$\alpha = 0.5$	$\beta = 0.4$	$\gamma = 0.3$	97.24
$\alpha = 0.5$	$\beta = 0.5$	$\gamma = 0.3$	97.02
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.2$	97.37
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.4$	97.47
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.5$	97.41

D. Module Comparison Experiments

In the module comparison experiments, we used three datasets, NWPU VHR-10, IPIU-M, and DOTA2.0, to validate the effectiveness of our proposed module, including the SAM as well as the CAM. We add each module to the SSD network in turn. The comparison of our method and SSD on the dataset with different categories of APs is shown in Tables XI–XIII. We show some detection comparison results in Figs. 9 and 10. The first column represents the detection of the backbone

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TABLE VIII
EXPERIMENTAL RESULTS OF HYPERPARAMETERS α, β, AND γ ON IPIU-M (%)

hyper-paramers			mAP
$\alpha = 0.4$	$\beta = 0.3$	$\gamma = 0.5$	76.79
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.5$	**77.27**
$\alpha = 0.6$	$\beta = 0.3$	$\gamma = 0.5$	76.82
$\alpha = 0.5$	$\beta = 0.2$	$\gamma = 0.5$	76.76
$\alpha = 0.5$	$\beta = 0.4$	$\gamma = 0.5$	76.70
$\alpha = 0.5$	$\beta = 0.5$	$\gamma = 0.5$	76.64
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.3$	76.47
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.4$	76.58
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.6$	76.56

TABLE IX
EXPERIMENTAL RESULTS OF HYPERPARAMETERS α, β, AND γ ON DIOR (%)

hyper-paramers			mAP
$\alpha = 0.4$	$\beta = 0.3$	$\gamma = 0.2$	78.3
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.2$	**78.9**
$\alpha = 0.6$	$\beta = 0.3$	$\gamma = 0.2$	78.4
$\alpha = 0.5$	$\beta = 0.2$	$\gamma = 0.2$	78.5
$\alpha = 0.5$	$\beta = 0.4$	$\gamma = 0.2$	78.4
$\alpha = 0.5$	$\beta = 0.5$	$\gamma = 0.2$	78.2
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.1$	78.4
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.3$	78.3
$\alpha = 0.5$	$\beta = 0.3$	$\gamma = 0.4$	78.1

TABLE X
EXPERIMENTAL RESULTS OF HYPERPARAMETERS α, β, AND γ ON DOTA2.0 (%)

hyper-paramers			mAP
$\alpha = 0.5$	$\beta = 0.4$	$\gamma = 0.3$	57.86
$\alpha = 0.6$	$\beta = 0.4$	$\gamma = 0.3$	**58.22**
$\alpha = 0.7$	$\beta = 0.4$	$\gamma = 0.3$	57.93
$\alpha = 0.6$	$\beta = 0.3$	$\gamma = 0.3$	58.05
$\alpha = 0.6$	$\beta = 0.5$	$\gamma = 0.3$	58.00
$\alpha = 0.6$	$\beta = 0.6$	$\gamma = 0.3$	57.91
$\alpha = 0.6$	$\beta = 0.4$	$\gamma = 0.1$	57.76
$\alpha = 0.6$	$\beta = 0.4$	$\gamma = 0.2$	57.88
$\alpha = 0.6$	$\beta = 0.4$	$\gamma = 0.4$	58.14

network and the right column represents the detection of our proposed method in this article.

1) Validation of the SAM Module: As shown in Table XI, on the dataset NWPU VHR-10, the overall mAP of the network with the addition of the SAM improved by 14.73 compared to SSD, and this phenomenon was evident for small objects. Taking Fig. 11 as an example, there are both large objects like ground track field and small objects like basketball court and tennis court in the original input image. We visualize and compare the feature maps of the same channel extracted by baseline and SSD ∪ SAM, we can see that the feature maps extracted by baseline are dominated by large objects, and the large strong responses in the training process will guide the network to put more attention to the area of large objects, resulting in the features of small objects being overwhelmed by large objects. However, the feature maps processed with the SAM can significantly suppress the high response of large objects in the low-level feature maps, thus allowing the network to focus more on the features of small objects in the low-level layer and improve the detection of small objects. Similarly in Tables XII and XIII, SAM also improves the mAP metrics in each category. From Table XIII, it can be seen that for categories such as SBF, RA, SP, and HC that contain objects of different scales, the AP value has increased by more than 10%. In addition, we also visually compare deep feature maps of the same channel in Fig. 12, and it is not difficult to find that the features of large objects are also enhanced in the deep feature maps after the SAM. Therefore, SAM not only suppresses the features of large objects in the low-level feature map and highlights the features of small objects, but also enhances the detailed features of large objects in the high-level feature map, making the feature information of large objects richer. It can be seen that the SAM module is beneficial to multiscale object detection.

2) Validation of the CAM Module: Similar to the above section, we added the CAM to the baseline network SSD, and the third row of Tables XI–XIII denotes the mAP of SSD ∪ CAM, respectively. It can be seen that our proposed module is quite consistently better than the baseline performance, and Fig. 13 shows the feature maps obtained before and after processing by the CAM. As we can see, the input image contains both small and dense objects like storage tank and ship-class objects. After the backbone extracted feature map (a), the features are confusing and unclear, and all the features extracted by the network are concentrated together. After the CAM, such as (b) and (c), (b) contains only the features of storage tank class objects, and (c) contains only the features of ship class objects, and the features of different classes of objects are separated into different channels to reduce the interclass feature interference, while blocking the background and other interference information, which well proves the effectiveness of our proposed module.

E. State-of-the-Art Methods

We have collected some advanced algorithms in the field of remote sensing object detection, including GACL [30], FAE-Net [31], Merge-Net [32], MSCNN [33], YOLOv8 [34], and GLSANet [6]. The following is a brief introduction to these algorithms.

GACL [30] proposes an axis-concentrated prediction module, which decouples the feature maps extracted by the network to different spatial directions to prevent feature interference between different directions. Specifically, it projects the feature maps into two spatial directions, x and y, by Max pooling operation to get the features p_x and p_y in two directions, which makes the information of the feature maps concentrated into 2 dimensions to predict (d_x, d_w) and (d_y, d_h) respectively, where d_x and d_y refer to the center coordinates of the predicted bounding box, and d_w and d_h refer to the width and height of the predicted bounding box. After decoupling the feature map into two independent spatial directions, although it can avoid the interference of features in different directions and improve the localization accuracy of small objects, it also ignores the spatial features of the image, which is not conducive to the detection of objects with relatively large length and width like bridge and harbor.

FAENet [31] enhances the semantic features of two adjacent low-level feature maps by adding feature aggregation modules

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TABLE XI
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON NWPU VHR-10 (%)

	airplane	baseball diamond	basketball court	bridge	ground track field	harbor	ship	storage tank	tennis court	vehicle	mAP
SSD	90.72	77.34	79.85	94.43	82.60	77.68	90.05	84.27	61.76	62.78	80.15
SSD$\cup SAM$	98.02	98.35	98.47	95.07	97.28	93.15	94.69	97.96	85.73	90.07	94.88
SSD$\cup CAM$	97.48	98.22	99.43	97.34	97.39	96.96	93.78	98.05	89.17	92.13	96.00
SSD$\cup SAM \cup CAM$	**98.41**	**100.0**	**99.78**	**98.21**	**98.33**	**97.54**	**96.95**	**99.89**	**93.58**	**94.30**	**97.70**

TABLE XII
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON IPIU-M (%)

	helicopter	Boeing	dense_ship	dense_ship	mAP
SSD	61.89	72.89	51.37	71.14	64.32
SSD$\cup SAM$	74.59	81.28	62.87	82.19	75.23
SSD$\cup CAM$	75.06	80.43	65.28	81.77	75.64
SSD$\cup SAM \cup CAM$	**76.85**	**81.67**	**66.09**	**84.46**	**77.27**

TABLE XIII
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON DOTA2.0 (%)

	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	CC	AP	HL	mAP
SSD	69.52	43.68	26.43	57.09	30.71	36.85	42.93	67.24	60.58	47.63	45.97	52.50	49.06	57.61	48.25	15.10	68.92	4.39	45.80
SSD$\cup SAM$	73.84	46.20	43.88	60.93	45.09	47.67	56.58	70.03	63.29	53.55	51.04	59.12	52.34	59.98	54.62	21.69	74.06	12.92	52.60
SSD$\cup CAM$	74.19	49.33	45.61	59.98	45.85	46.29	58.17	71.21	65.02	53.82	53.15	57.81	56.86	64.15	53.67	23.99	76.41	14.64	53.90
SSD$\cup SAM \cup CAM$	**76.93**	**51.75**	**47.00**	**63.95**	**52.38**	**54.16**	**66.82**	**73.01**	**68.91**	**59.75**	**58.86**	**63.42**	**59.72**	**66.09**	**58.96**	**27.38**	**80.16**	**18.77**	**58.22**

and feature enhancement blocks, respectively. This operation significantly enhances the detailed texture features of the airplane, greatly improving the detection performance of the airplane. However, FAENet ignores the feature interaction between the low-level and high-level feature maps. The lack of detailed texture features of the object in the deep feature map results in low positioning accuracy, and the design of the low-level FEM increases the depth of the network, making it easier for small object features to be lost during forward propagation.

Merge-Net [32] fuses the detailed features of the low-level feature map and the semantic features of the deep feature map by introducing a top-down connection. Specifically, the deep feature map is upsampled by a deconvolutional layer and then fused with the previous feature map through a lateral connection.

MSCNN [33] network expands the perceptual field by introducing dilated bottleneck structure in the backbone to obtain richer contextual features, especially for the bridge category with relatively large length and width, the contextual features needed to identify bridges are extracted by using dilated convolution. Therefore, MSCNN has more obvious effect for the bridge category.

YOLOv8 [34] draws on the design advantages of models such as Yolov5, Yolov6, and YoloX, comprehensively improving the structure of Yolov5 models while maintaining the advantages of engineering simplicity and ease of use. The backbone section of YOLOv8 has replaced all C3 modules with C2f with richer gradient flow. The neck section retains the PAN idea and removes the convolutional structure in the upsampling stage. The head section uses the current mainstream decoupling head structure. YOLOv8 uses VFL loss as classification loss and DFL loss and CIOU loss as regression loss. VFL loss is designed to address the extreme imbalance between foreground and background classes in dense object detector training. DFL loss is designed to address the issues of imbalanced object categories and imbalanced difficult samples.

GLSANet [6] proposed a global-to-local scale-aware detection network, designed a global semantic information interaction module, and introduced a local attention pyramid. The global semantic information interaction module is mainly used to mine and enhance advanced semantic information in deep feature maps. The local attention pyramid module mainly enhances the features of small objects in each channel, while suppressing the background and noise, effectively solving the problem of complex background interference and a large number of small objects in remote sensing images.

CoF-Net [35] proposes a new coarse-to-fine remote sensing image object detection framework, which mainly consists of two parallel branches: coarse-to-fine feature adaptation (CoF-FA) and coarse-to-fine sample allocation (CoF-SA). CoF-FA smoothly refines the original coarse features into multispectral nonlocal fine features with discriminative spatial–spectral details and semantic relationships. CoF-FA smoothly refines the original coarse features into multispectral nonlocal fine features that possess discriminative spatial-spectral details and semantic relationships. CoF-Net effectively mitigates the problem of rough features and samples caused by complex background interference, dense and chaotic instance arrangement, and large-scale changes in remote sensing object detection.

SB-MSN [36] proposes a new multilevel network based on sampling balance, which uses a multiscale information

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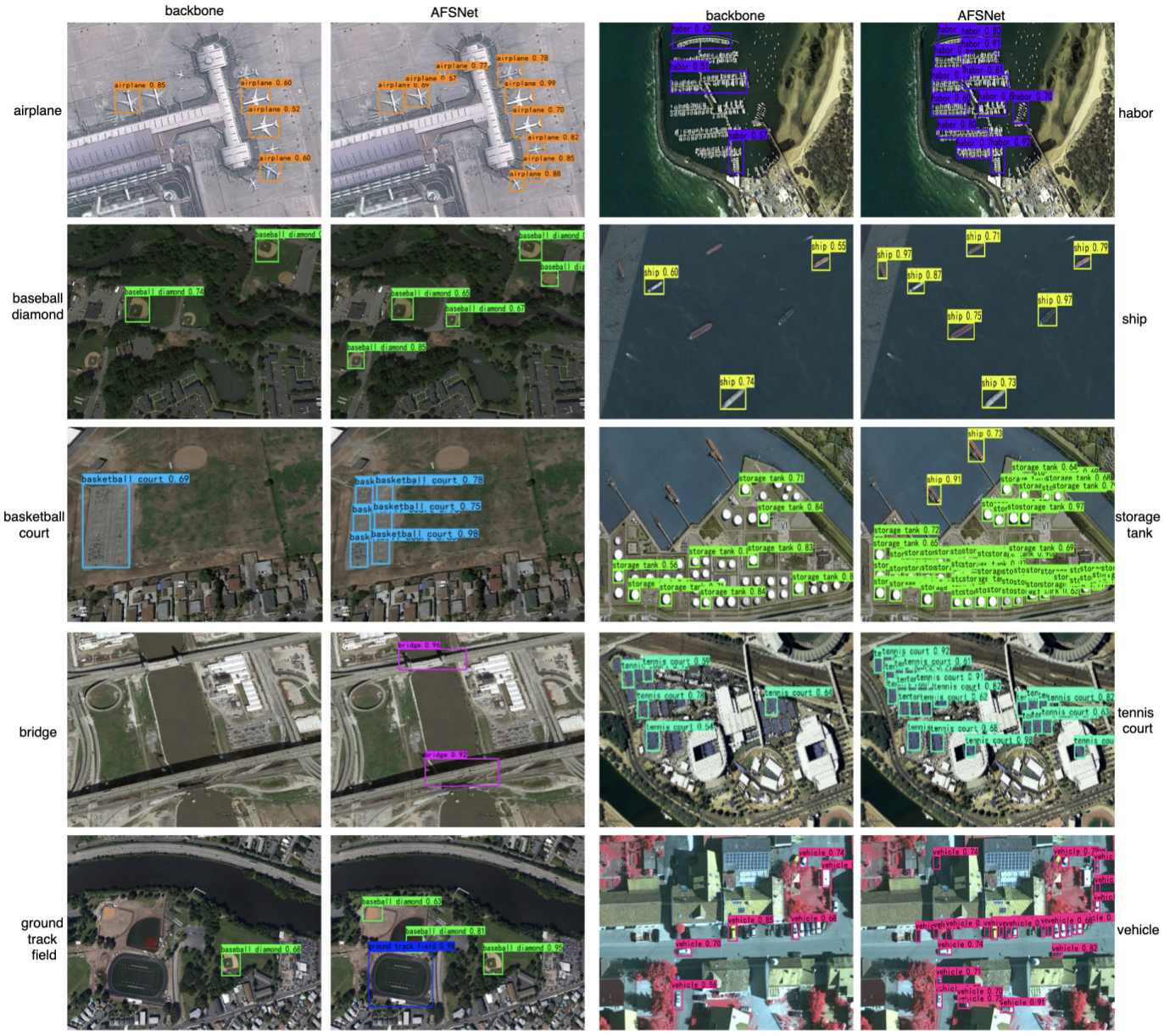


Fig. 9. Detection results comparison between the backbone network and the proposed AFSNet on NWPU VHR-10.

preservation module to construct a series of pyramid images from high resolution to low resolution based on predefined rules, and then applies a feature extractor to learn and generate multiscale features. Second, SB-MSN adopts an IoU balanced sampling strategy to adaptively select high-quality negative sampling for subsequent training. Finally, SB-MSN replaced smooth $L1$ loss with balanced $L1$ loss during the regression process, significantly reducing the impact of unbounded regression objects during training.

RSADet [37] proposes a novel remote sensing image object detection framework that takes into account the unique spatial distribution and other spatial characteristics of remote sensing objects. The first stage of RSADet infers each object instance on the CNN heatmap with increased attention to generate candidate bounding boxes. In the second stage, RSADet introduces deformable convolution to adapt to the geometric changes of different object instances and avoid the influence of complex and variable backgrounds. In addition, RSADet introduces a new bounding box confidence (IoU score) prediction branch, effectively eliminating unreliable prediction boxes.

LGDA [38] has proposed a small object detection pipeline. LGDA introduces a partial crossover stage for low/high resolution aggregation and replaces maximum pooling with spatial pyramid pooling to achieve finer feature extraction. Second, the difficulty estimator module proposed by LGDA calculates the overall difficulty of each image as a weight parameter in the focal loss function to improve object localization and recognition. Finally, LGDA also added two progressive domain adaptive blocks using contrastive learning for local feature alignment and global feature alignment.

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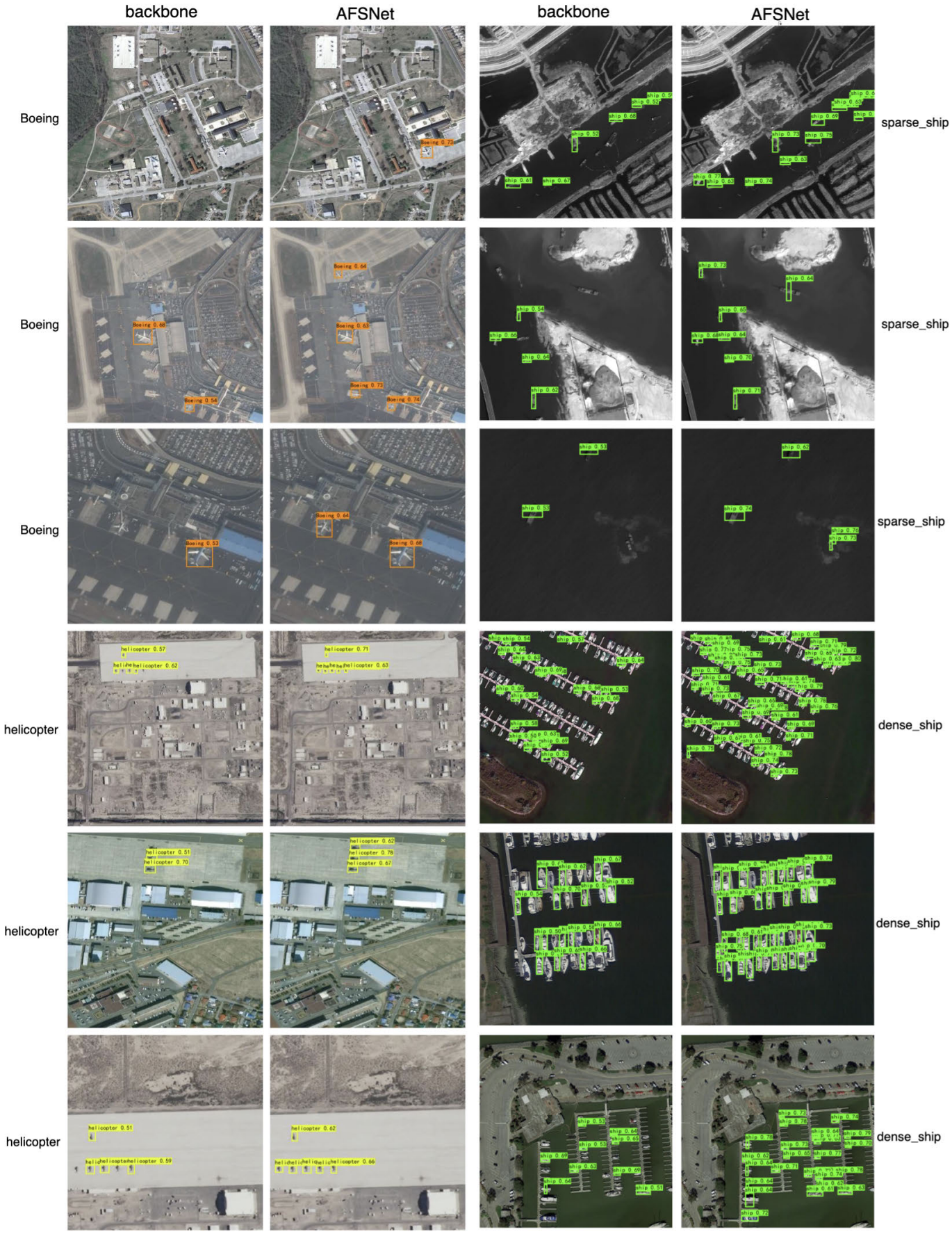


Fig. 10. Detection results comparison between the backbone network and the proposed AFSNet on IPIU-M.

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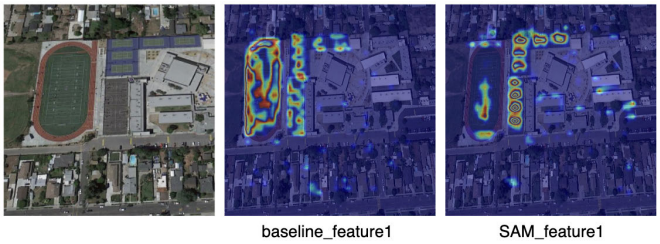


Fig. 11. Comparison of the first predicted feature map obtained between baseline and SAM module in the same channel. The feature map on the left is extracted from the baseline, and the features of small objects in this feature map are easily drowned out by the strong response of adjacent large objects. The right feature map is after passing through the SAM module. In this feature map, the features of large objects are suppressed, while the features of small objects are highlighted, which is more conducive to shallow detection of small objects.

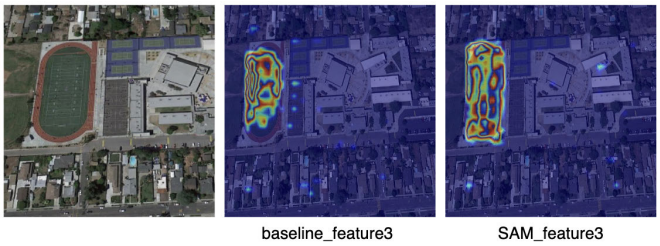


Fig. 12. Comparison of the third predicted feature map obtained between baseline and SAM module in the same channel. Compared with the left feature map extracted from the baseline, the right feature map after passing through the SAM module has significantly richer features for large objects.

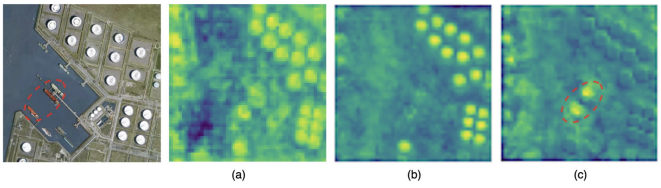


Fig. 13. (a) Feature map that has not been processed by the CAM module, with fuzzy features and unclear object boundaries. (b) and (c) Feature maps processed by the CAM module, and features of the storage tank class are separated into one channel, as shown in (b), and the boundary between the objects of the storage tank in (b) is clearer than in (a). Features of the ship class are separated into another channel, as shown in (c). The objects' features are clearer.

F. Comparison With Other Methods

To evaluate the performance of our proposed method in this article, we compare it with several existing state-of-the-art (SOTA) methods as discussed in Section IV-E. We conducted experiments on each of these methods on our dataset, and the experimental results are shown in Tables XIV–XVI.

Tables XIV–XVI demonstrate the effectiveness of our proposed method in this article, which achieves the best mAP among all the other methods mentioned in the comparison. In the NWPU experiments, our detection results are significantly improved, especially on small and dense categories, such as small and dense storage tank, ship with generally small sizes, and vehicle. In the comparison experiments of IPIU-M, our proposed model achieves higher detection results than other models on all categories. In the comparative experiments on the DIOR dataset, our algorithm also achieved competitive results, achieving the best results in 9 out of a total of 20 categories.

It can be seen from Table XV that GLASNet has a certain degree of robustness in small object detection, such as achieving high mAP in small object categories such as baseball fields (99.41%) and tennis courts (98.34%). However, it performs poorly in detecting vehicles, achieving only 87.08% AP in this category. This may be because the image containing the vehicle is a locally sharpened color infrared image, rather than typically captured by satellites. Therefore, GLSANet may not be able to adapt to these infrared images, resulting in poor performance in this category. CoF-Net achieved 100.0% mAP in the airplane category and the highest mAP in the ground track field category (99.70%). SB-MSN also effectively mitigates the scale difference problem in remote sensing object detection, achieving the maximum mAP in airplane, basketball court, tennis court, and vehicle categories. Our AFSNet achieved the best accuracy with an overall mAP of 97.7%. In terms of single-category AP, AFSNet achieved the best AP in five out of ten categories, including baseball diamond (100.0%), bridge (98.21%), harbor (97.54%), and storage tank (99.89%). Among them, the baseball diamond category contains a large number of small objects, indicating that our method can detect objects well at different scales. The optimal performance of bridge and storage tank demonstrates the effectiveness and superiority of our method in detecting objects similar to the background and densely arranged objects.

It can be seen from Table XVI that compared to RSADet and SB-MSN algorithms, GLSANet and CoF-Det perform better on the DIOR dataset, with mAP values of 77.9% and 75.77%, respectively. Specifically, GLSANet achieved the best results in many categories. Our method AFSNet is slightly better than GLSANet, achieving a detection accuracy of 78.9% on the overall mAP, an improvement of 1% compared to GLSANet, and an improvement of 3.13% compared to CoF-Det. AFSNet achieved the best results in 8 out of 20 categories in the DIOR dataset. Specifically, for example, BF or BC, their distribution is more dense and complex. Like WM and ES, their characteristics are very weak and not obvious. These characteristics place high demands on the detection model. Although our AFSNet has achieved good detection results for multiclass objects in the dataset, proving its effectiveness, there is still a lot of room for improvement. In the future, we need to do more work to further improve the recognition performance of the DIOR dataset.

G. Ablation Study

To verify the importance of each module, we gradually added the SAM and the class-aware module to the baseline network SSD. Tables XVII and XVIII show the results of the ablation experiments on two different datasets.

1) SAM Module: As shown in Tables XVII and XVIII, the mAP increases significantly after adding the SAM, which indicates that the SAM can make the network scale robust and can better capture object features at different scales, thus enabling multiscale object detection.

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TABLE XIV
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON IPIU-M (%)

	helicopter	Boeing	dense_ship	sparse_ship	mAP
SSD [39]	61.89	72.89	51.37	71.14	64.32
GACL [30]	70.66	77.92	59.89	79.14	71.90
FAE-Net [31]	62.18	64.69	56.43	73.82	64.28
Merge-Net [32]	64.32	75.01	54.60	74.47	67.10
MSCNN [33]	69.69	73.04	57.02	63.58	65.83
AFSNet(ours)	**76.85**	**81.67**	**66.09**	**84.46**	**77.27**

TABLE XV
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON NWPU VHR-10 (%)

	airplane	baseball diamond	basketball court	bridge	ground track field	harbor	ship	storage tank	tennis court	vehicle	mAP
YOLOv8 [34]	99.88	98.07	91.43	84.53	96.66	91.12	93.7	96.59	94.37	90.65	93.7
GLSANet [6]	99.86	99.41	86.17	84.89	99.43	97.47	95.60	97.04	98.34	87.08	94.53
CoF-Net [35]	**100.0**	98.85	95.73	89.58	**99.70**	92.05	90.62	96.39	91.54	90.60	94.51
SB-MSN [36]	**100.0**	98.45	**99.82**	92.51	95.39	88.56	95.43	94.19	**98.77**	**98.43**	96.16
LGDA [38]	98.75	98.43	94.62	89.81	94.17	93.05	91.28	95.70	96.93	88.02	94.08
AFSNet(ours)	98.41	**100.0**	99.78	**98.21**	98.33	**97.54**	**96.95**	**99.89**	93.58	94.30	**97.70**

TABLE XVI
COMPARISON OF OBJECT DETECTION WITH OTHER METHODS ON DIOR (%)

	AL	AT	BF	BC	BG	CM	DM	EA	ES	GC	GF	HB	OP	SP	SD	ST	TC	TS	VH	WM	mAP
RSADet [37]	73.6	**86.0**	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	57.9	59.2	90.0	55.8	77.0	87.8	65.3	55.3	86.5	72.2
GLSANet [6]	**95.8**	78.9	92.9	87.9	50.7	81.1	55.5	79.8	74.1	71.6	**87.6**	**66.4**	**65.5**	**95.2**	**92.4**	**86.3**	**94.8**	50.6	**62.1**	89.2	77.9
CoF-Net [35]	84.0	85.3	82.6	90	47.1	80.7	73.3	**89.3**	74.0	84.5	83.2	57.4	62.2	82.9	77.6	68.2	89.9	**68.7**	49.3	85.2	75.77
SB-MSN [36]	79.6	82.2	76.4	89.8	45.6	78.2	64.8	58.9	59.3	79.2	82.4	51.8	60.8	74.4	79.7	66.4	85.6	65.4	45.1	79.9	70.3
LGDA [38]	79.6	83.4	81.8	89.2	52.8	73.7	68.1	84.5	61.2	77.3	81.9	60.1	55.2	89.7	64.9	82.7	85.2	63.9	58.5	88.6	74.1
AFSNet(ours)	91.6	83.4	**96.1**	**94.9**	**56.2**	**87.6**	**79.4**	64.6	**76.8**	**87.9**	73.3	60.5	61.7	94.3	87.4	77.4	89.2	65.7	57.9	**92.9**	**78.9**

TABLE XVII
EXPERIMENTAL RESULTS OF EACH COMPONENT ON NWPU VHR-10 VALIDATION (%)

SAM	CAM	airplane	baseball diamond	basketball court	bridge	ground track field	harbor	ship	storage tank	tennis court	vehicle	mAP
✓		98.02	98.35	98.47	95.07	97.28	93.15	94.69	97.96	85.73	90.07	94.88
✓	✓	**98.41**	**100.0**	**99.78**	**98.21**	**98.33**	**97.54**	**96.95**	**99.89**	**93.58**	**94.30**	**97.70**

TABLE XVIII
EXPERIMENTAL RESULTS OF EACH COMPONENT ON IPIU-M VALIDATION (%)

SAM	CAM	helicopter	Boeing	dense_ship	sparse_ship	mAP
✓		74.59	81.28	62.87	82.19	75.23
✓	✓	**76.85**	**81.67**	**66.09**	**84.46**	**77.27**

2) CAM Module: Adding the CAM module on top of the SAM module in the baseline, the mAP is improved from 94.88 to 97.70 and from 75.23 to 77.27 in Tables XVII and XVIII, respectively. Especially on the NWPU VHR-10 dataset, the mAP of vehicle category containing a large number of small objects has been improved by more than 3, because the feature maps extracted from the backbone include features of various categories and complex background features. Small objects usually contain fewer and densely arranged features, which are easily disturbed by other categories and submerged by the background. The class-aware module separates confusing features by category, effectively reducing category interference and background influence.

V. CONCLUSION

From the perspective of optimizing feature maps, this article proposes an AFSNet. The core idea of the network is to decouple the complex feature maps extracted from the baseline network into different channels in proportion and category. Feature maps of different resolutions and different channels of each feature map can be clearly assigned, and features do not interfere with each other for better detection. Through experiments, it has been proven that our SAM can effectively suppress the strong response of large objects in low-level feature maps, highlight the features of small objects, and prevent them from being submerged by large objects. The class-aware module can decouple complex feature maps by

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category in the channel dimension, effectively alleviating the problems of interclass interference and background interference. On the NWPU VHR-10 dataset, our model achieved a 27% improvement compared to the baseline model, and a nearly 28% improvement in mAP performance on a single category of vehicles containing a large number of small objects, achieving the highest mAP performance compared to other algorithms. On the IPIU-M dataset, our model improved by about 13% compared to the SSD algorithm, and for dense ship categories, our algorithm improved by nearly 15%. On the DOTA2.0 dataset, our model improved overall mAP by 13% compared to the baseline model SSD, improved mAP by nearly 23% on single-category small vehicles containing a large number of small objects, and improved mAP by nearly 24% on ship categories similar to the background. This work can provide some reference ideas for solving the problem of poor detection performance of small and dense weak objects in remote sensing images. However, due to the optimization of each layer's feature map in our network, the computational complexity is higher than that of the baseline network, and the number of parameters also increases significantly. Moreover, for different datasets, we need to adjust the weights of each part of the loss function, which to some extent increases the experimental complexity. Afterward, we will continue to optimize and propose simpler and more effective methods to reduce model complexity and improve model robustness.

REFERENCES

[1] S. Chang, B. Du, and L. Zhang, "A subspace selection-based discriminative forest method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4033–4046, Jun. 2020.

[2] L. Shutao, W. Qiong, and K. Xudong, "Hyperspectral remote sensing image intrinsic information decomposition: Advances and challenges," *Acta Geodaetica et Cartographica Sinica*, vol. 52, no. 7, pp. 1059–1073, 2023.

[3] M. Zhao, W. Li, L. Li, P. Ma, Z. Cai, and R. Tao, "Three-order tensor creation and Tucker decomposition for infrared small-target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2021.

[4] Y. Liu, Z. Xiong, Y. Yuan, and Q. Wang, "Distilling knowledge from super-resolution for efficient remote sensing salient object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609116.

[5] V. Chalavadi, P. Jeripothula, R. Datla, S. B. Ch, and C. K. Mohan, "MSODANet: A network for multi-scale object detection in aerial images using hierarchical dilated convolutions," *Pattern Recognit.*, vol. 126, Jul. 2022, Art. no. 108548.

[6] T. Gao, Q. Niu, J. Zhang, T. Chen, S. Mei, and A. Jubair, "Global to local: A scale-aware network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615614.

[7] S. Cho, T. J. Jun, B. Oh, and D. Kim, "DAPAS: Denoising autoencoder to prevent adversarial attack in semantic segmentation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2020, pp. 1–8.

[8] C. Tian, Y. Xu, Z. Li, W. Zuo, L. Fei, and H. Liu, "Attention-guided CNN for image denoising," *Neural Netw.*, vol. 124, pp. 117–129, Apr. 2020.

[9] C. Xie, Y. Wu, L. van der Maaten, A. L. Yuille, and K. He, "Feature denoising for improving adversarial robustness," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 501–509.

[10] Z. Liu, G. Gao, L. Sun, and L. Fang, "IPG-Net: Image pyramid guidance network for small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 4422–4430.

[11] C. Deng, M. Wang, L. Liu, Y. Liu, and Y. Jiang, "Extended feature pyramid network for small object detection," *IEEE Trans. Multimedia*, vol. 24, pp. 1968–1979, 2022.

[12] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[13] Q. Wang, Y. Liu, Z. Xiong, and Y. Yuan, "Hybrid feature aligned network for salient object detection in optical remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5624915.

[14] Z. Huang, W. Li, X.-G. Xia, X. Wu, Z. Cai, and R. Tao, "A novel nonlocal-aware pyramid and multiscale multitask refinement detector for object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5601920.

[15] W. Ma et al., "Feature split—merge–enhancement network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5616217.

[16] H. Fu et al., "LRAF-Net: Long-range attention fusion network for visible–infrared object detection," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Jun. 6, 2023, doi: 10.1109/TNNLS.2023.3266452.

[17] X. Li et al., "A complex-former tracker with dynamic polar spatio-temporal encoding," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Aug. 21, 2024, doi: 10.1109/TNNLS.2023.3302368.

[18] Y. Liu, X.-Y. Zhang, J.-W. Bian, L. Zhang, and M.-M. Cheng, "SAMNet: Stereoscopically attentive multi-scale network for lightweight salient object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 3804–3814, 2021.

[19] X. Li et al., "A collaborative learning tracking network for remote sensing videos," *IEEE Trans. Cybern.*, vol. 53, no. 3, pp. 1954–1967, Mar. 2023.

[20] C. Wu, B. Du, and L. Zhang, "Fully convolutional change detection framework with generative adversarial network for unsupervised, weakly supervised and regional supervised change detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9774–9788, Aug. 2023.

[21] Y. Zhu, X. Sun, W. Diao, H. Li, and K. Fu, "RFA-Net: Reconstructed feature alignment network for domain adaptation object detection in remote sensing imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 5689–5703, 2022.

[22] D. Biswas and J. Tešič, "Unsupervised domain adaptation with debiased contrastive learning and support-set guided pseudolabeling for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3197–3210, 2024.

[23] J. Lee, H. Kim, C. Park, J. Jang, and J. Paik, "Small object detection in infrared images using attention mechanism and sigmoid function," in *Proc. IEEE Int. Conf. Consum. Electron. (ICCE)*, Jan. 2024, pp. 1–3.

[24] M. Lei and X. Liu, "SOLO-Net: A sparser but wiser method for small object detection in remote-sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[25] X. Han, Y. Wang, Q. Wang, N. Zhang, and C. Liu, "Vertical and horizontal network for small object detection in sports videos," in *Proc. IEEE 7th Adv. Inf. Technol., Electron. Autom. Control Conf. (IAEAC)*, vol. 7, Mar. 2024, pp. 1879–1884.

[26] Q. Wei, X. Jiang, Y. Liu, Q. Su, and M. Yu, "Small object detection on the water surface based on radar and camera fusion," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Apr. 2024, pp. 8381–8385.

[27] Y. Gao, Y. Wang, Y. Zhang, Z. Li, C. Chen, and H. Feng, "Feature super-resolution fusion with cross-scale distillation for small-object detection in optical remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[28] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[29] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[30] X. Lu, Y. Zhang, Y. Yuan, and Y. Feng, "Gated and axis-concentrated localization network for remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 179–192, Jan. 2020.

[31] W. Li and G. Liu, "A single-shot object detector with feature aggregation and enhancement," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2019, pp. 3910–3914.

[32] P. Wang, X. Sun, W. Diao, and K. Fu, "MergeNet: Feature-merged network for multi-scale object detection in remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2019, pp. 238–241.

[33] Q. Yao, X. Hu, and H. Lei, "Multiscale convolutional neural networks for geospatial object detection in VHR satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 1, pp. 23–27, Jan. 2021.

[34] G. Jocher, A. Chaurasia, and J. Qiu. (Jan. 2023). *Ultralytics YOLO*. [Online]. Available: https://github.com/ultralytics/ultralytics

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[35] C. Zhang, K.-M. Lam, and Q. Wang, "CoF-Net: A progressive coarse-to-fine framework for object detection in remote-sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600617.
[36] W. Han et al., "Improving training instance quality in aerial image object detection with a sampling-balance-based multistage network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10575–10589, Dec. 2021.
[37] D. Yu and S. Ji, "A new spatial-oriented object detection framework for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4407416.
[38] D. Biswas and J. Tešič, "Domain adaptation with contrastive learning for object detection in satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5620615.
[39] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. 14th Eur. Conf. Comput. Vision (ECCV)*. Amsterdam, The Netherlands: Springer, Oct. 2016, pp. 21–37.



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